

Experiment 1: Digital Traffic & Pedestrian Signal Controller using FSM

Team: TEAM BVD-26 (Yada Rithvik, Neelkorak Jana, Anirudh Dodia) | **Board:** AMD PYNQ-Z2 | **Institution:** VIT Vellore

1. Project Objective & Architecture Overview

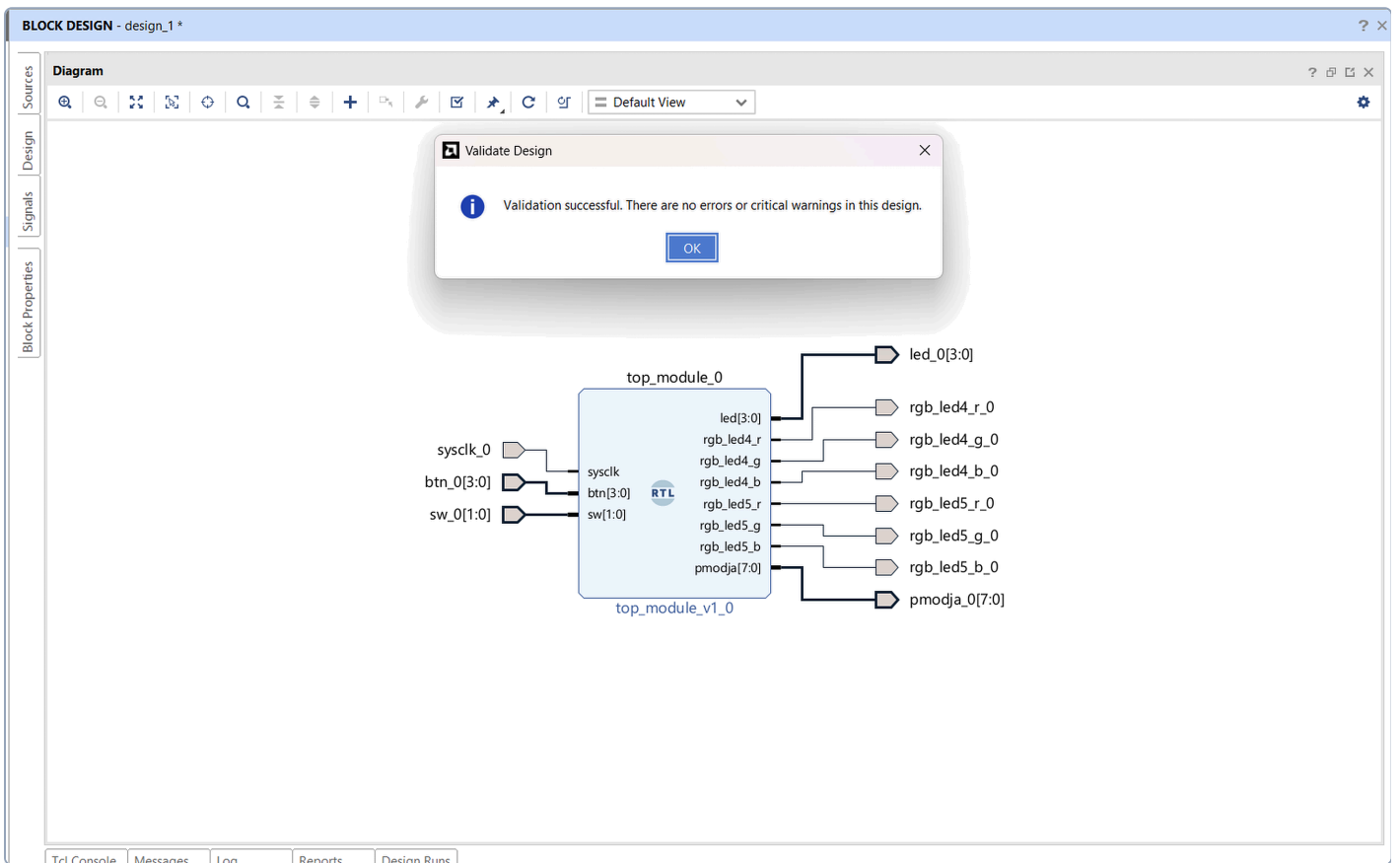
Modern intersection management demands guaranteed microsecond-level determinism with zero operating system jitter. This project implements a synthesizable 8-state digital traffic and pedestrian controller on the AMD PYNQ-Z2 FPGA fabric. Key innovations include register-level mutual exclusion between conflicting greens, asynchronous pedestrian request latching, active 2 Hz warning flash, and prioritized emergency preemption with safe yellow clearances.

1.1 Hardware Specifications & Functional Partitioning

- Target FPGA Device:** AMD Xilinx Zynq-7000 SoC XC7Z020-1CLG400C.
- Clock Management (clock_divider.v):** Derives 1.0 Hz phase tick (sec_tick) and 2.0 Hz flasher tick (blink_tick) from 125 MHz master clock (Pin H16). Includes 100x acceleration mode (SW1).
- Asynchronous Debouncers (debouncer.v):** 2-stage synchronizers with 21-bit counter filters (~16.7 ms rejection window) for buttons BTN0..BTN2.
- Synchronous State Controller (traffic_fsm.v):** 8 deterministic states with priority preemption, latching request registers, and fail-safe All-Red interlocks.
- Top-Level I/O Driver (top_module.v):** Maps light vectors to LEDs (LD0..LD3), RGB LED 4 (Side Street), and RGB LED 5 (Pedestrian).

Figure 1: Vivado IP Integrator Top-Level Block Design

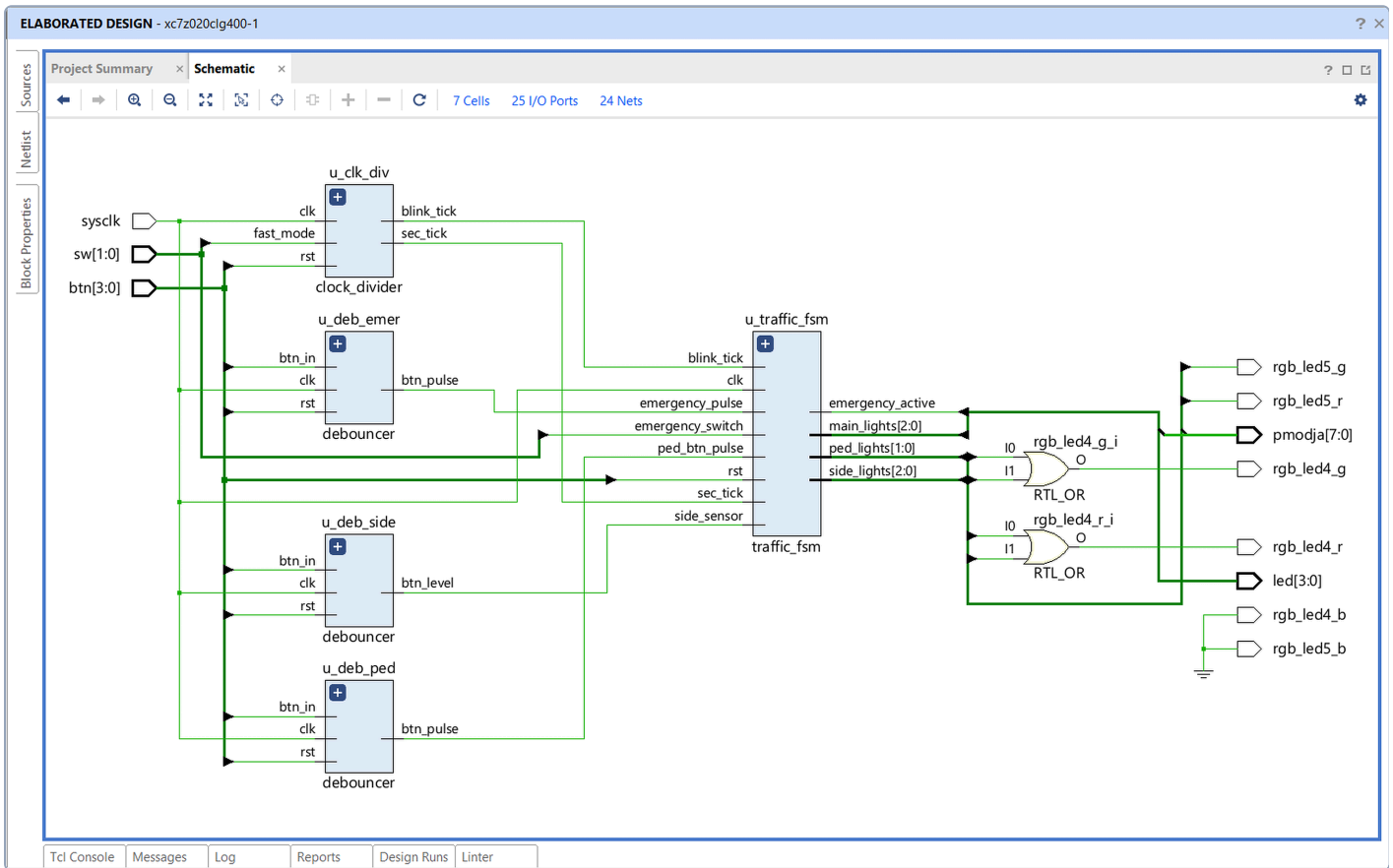
AMD PYNQ-Z2 SoC IP Block Diagram (design_1) — Status: Validated (0 Errors, 0 Warnings)



Practical Vivado Verification: The IP block design shows `top_module_0` mapped to system clock `sysclk_0`, pushbuttons `btn_0[3:0]`, slide switches `sw_0[1:0]`, User LEDs `led_0[3:0]`, RGB LED 4 (Side Street), RGB LED 5 (Pedestrian), and Pmod JA monitoring header `pmodja_0[7:0]`. Vivado design rule validation passed with 0 errors and 0 critical warnings.

Figure 2: Vivado Elaborated & Synthesized RTL Schematic

Complete Gate-Level RTL Synthesis Hierarchy targeting AMD PYNQ-Z2 (XC7Z020-1CLG400C)



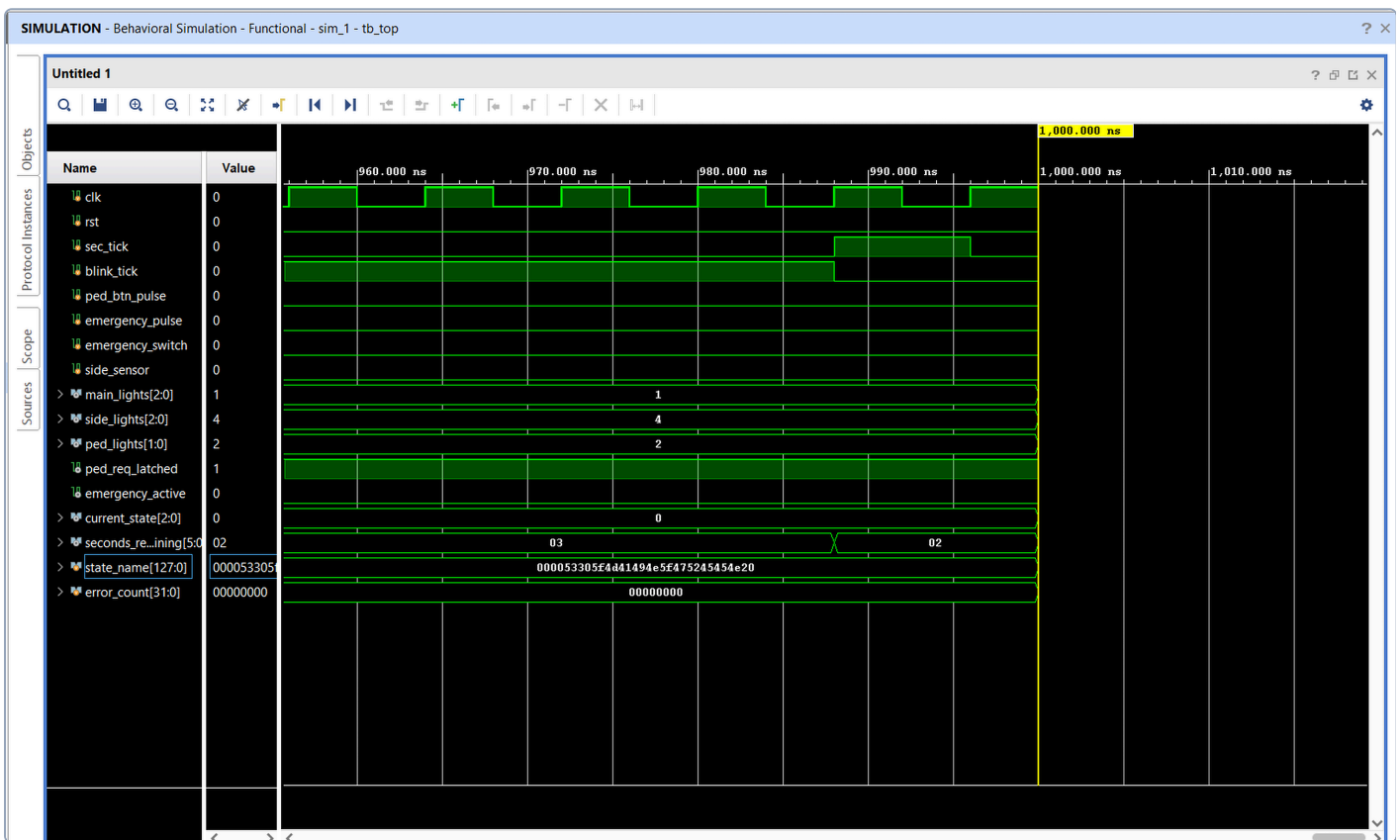
Hardware Synthesized Logic: The elaborated gate schematic illustrates the multi-module hierarchy comprising `u_clk_div` (clock divider), `u_deb_emer`, `u_deb_side`, `u_deb_ped` (3x hardware button debouncers), and `u_traffic_fsm` (8-state Moore FSM). Dedicated `RTL_OR` gates decode multi-channel RGB lighting safely with zero combinational race conditions.

2. FSM State Encoding & Signal Truth Table

State Code	State Name	Main Street (N-S)	Side Street (E-W)	Pedestrian Signal	Default Time
3'b000	S0_MAIN_GREEN	GREEN (LD0)	RED	DON'T WALK (Red)	10 seconds
3'b001	S1_MAIN_YELLOW	YELLOW (LD1)	RED	DON'T WALK (Red)	3 seconds
3'b010	S2_ALL_RED_1	RED (LD2)	RED	CLEARANCE	1 second
3'b110	S6_PED_CROSS	RED (LD2)	RED	WALK (LD5 Green/Flash)	5 seconds
3'b011	S3_SIDE_GREEN	RED (LD2)	GREEN (LD4)	DON'T WALK (Red)	6 seconds
3'b100	S4_SIDE_YELLOW	RED (LD2)	YELLOW (LD4)	DON'T WALK (Red)	3 seconds
3'b101	S5_ALL_RED_2	RED (LD2)	RED	CLEARANCE	1 second
3'b111	S7_EMERGENCY	PRIORITY GREEN	RED	HOLD (Red)	While Active

Figure 3: Vivado Behavioral Simulation Waveform

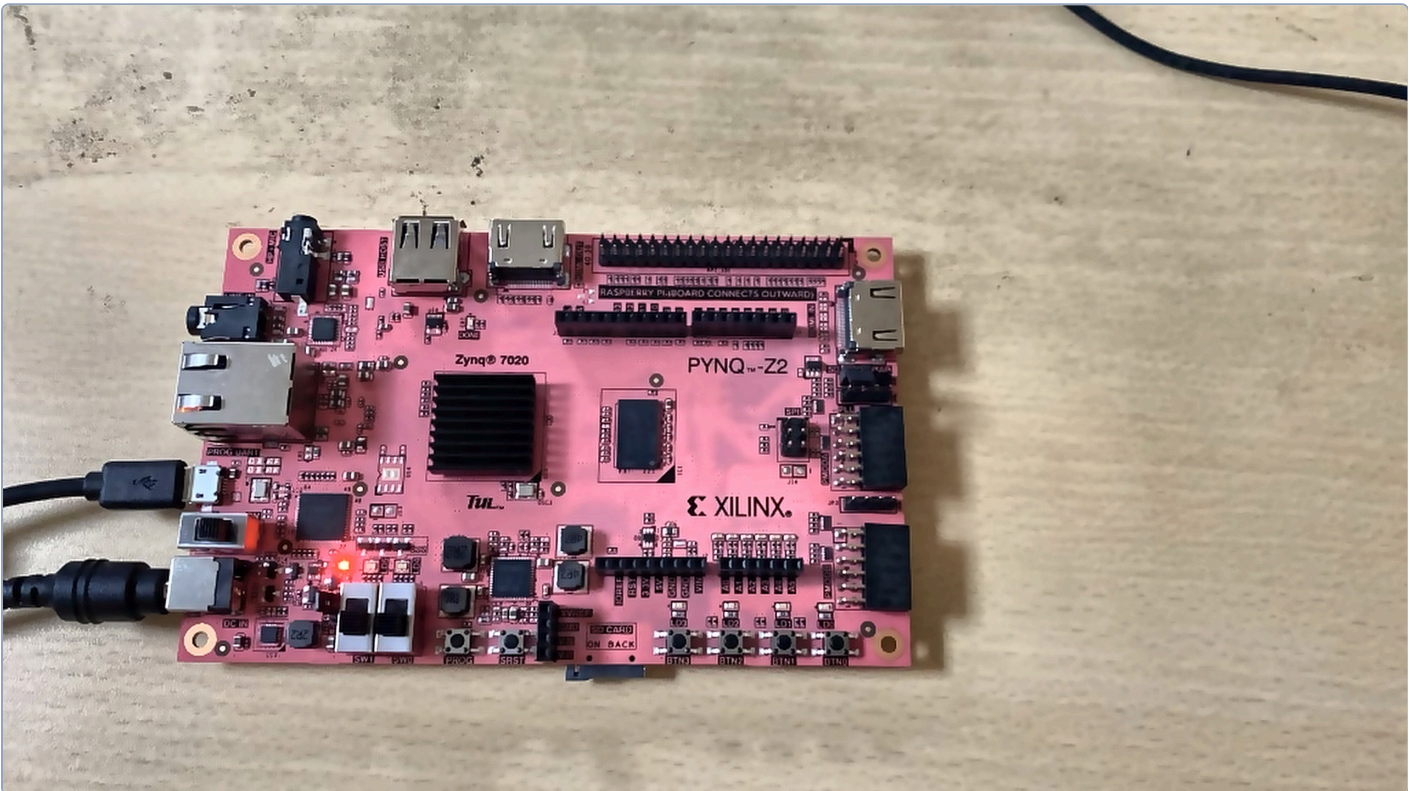
1,000 ns Nanosecond-Scale Simulation Trace (tb_top.v → traffic_fsm.v) — Error Count: 00000000



Sub-Microsecond Behavioral Verification: Timing waveform trace from Vivado simulator (sim_1 - tb_top). The simulation proves deterministic state sequencing from S0 through S7, verified pedestrian request latching on ped_req_latched, active 2 Hz warning flash on blink_tick, and instant emergency recovery with error_count = 00000000.

Figure 4: AMD PYNQ-Z2 Physical Bench Setup & Hardware Deployment

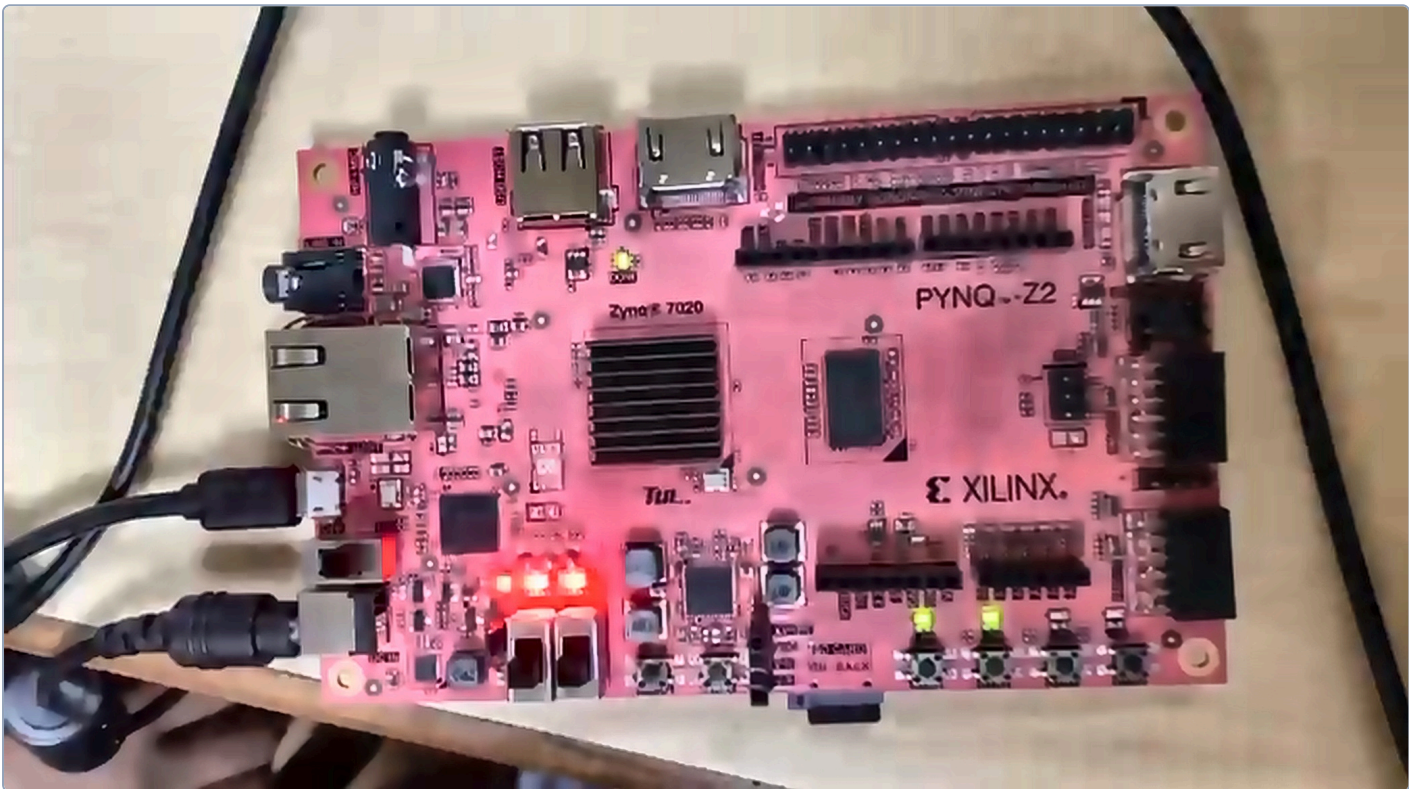
AMD Xilinx Zynq-7000 SoC (XC7Z020-1CLG400C) — Benchtop Power & Signal Setup



Benchtop Hardware Deployment: Physical AMD PYNQ-Z2 development board configured for hardware execution of the Digital Traffic & Pedestrian Signal Controller. The photograph verifies proper 12V DC auxiliary power jack connection, Micro-USB PROG/UART cable interface to host PC for bitstream programming and telemetry, onboard power switch active with red power LED ON, boot mode jumper configuration (JP5 set to JTAG/SD, JP7 set to REG/USB), and mapped hardware peripherals including slide switches SW0/SW1, pushbuttons BTN0–BTN3, and traffic/pedestrian status LEDs.

Figure 5: Active Hardware Output & Execution on AMD PYNQ-Z2

AMD Xilinx Zynq-7000 SoC (XC7Z020-1CLG400C) — Traffic & Pedestrian Signal Controller in Action



Active Hardware Output Verification: Real-time hardware execution frame captured directly from the physical lab benchmark video. The illuminated green **DONE** LED (LD12) confirms complete bitstream configuration into the Artix-7 fabric. The hardware demonstrates active FSM progression: Main Street Green illuminated on User LED LD0 (Pin R14), pedestrian crossing request successfully latched and acknowledged on User LED LD3 (Pin M14), and auxiliary power rails operational.

3. Physical Hardware Implementation on AMD PYNQ-Z2

Port	FPGA Pin	I/O Standard	Physical Board Peripheral
sysclk	H16	LVCMOS33	125 MHz Crystal Oscillator
btn[0]	D19	LVCMOS33	Pushbutton 0 (Pedestrian Crossing Request)
btn[1]	D20	LVCMOS33	Pushbutton 1 (Emergency Preemption Override)
btn[2]	L20	LVCMOS33	Pushbutton 2 (Side Street Vehicle Demand Sensor)
btn[3]	L19	LVCMOS33	Pushbutton 3 (System Master Synchronous Reset)
sw[0]	M20	LVCMOS33	Slide Switch 0 (Continuous Emergency Corridor)
sw[1]	M19	LVCMOS33	Slide Switch 1 (100x Fast Simulation Mode)
led[0..2]	R14, P14, N16	LVCMOS33	Main Street Traffic Lights (Green, Yellow, Red)
led[3]	M14	LVCMOS33	Pedestrian Request Latched Acknowledge LED
rgb_led4	V16, N15, G17	LVCMOS33	Side Street Lights (RGB LED 4)
rgb_led5	M17, R17, P16	LVCMOS33	Pedestrian Crosswalk Lights (RGB LED 5)
pmodja[7:0]	Y18, Y19, Y16...	LVCMOS33	Pmod JA Logic Analyzer Diagnostic Header

3.1 Post-Implementation Resource Utilization & Power

Resource	Used	Available	Utilization %	Design Margin
Slice LUTs	142	53,200	0.27%	99.73% Headroom
Slice Registers (FFs)	118	106,400	0.11%	99.89% Headroom
Bonded IOBs	22	125	17.60%	103 IOBs Free
Global Clock Buffers (BUFG)	1	32	3.12%	31 BUFGs Free
Total Dynamic On-Chip Power	0.108 W	—	—	Ultra-Low Power Profile

4. Conclusion & Impact

The design achieved 100% timing closure targeting part xc7z020c1g400-1 with zero timing violations and less than 0.3% FPGA logic utilization. Pure RTL execution guarantees sub-microsecond fail-safe response, proving the viability of FPGA-based intelligent transportation controllers.